Picklist App – Product Assignment

Problem Understanding

Overview:

- Describe the current challenges in warehouse order picking.
- Highlight inefficiencies, errors, and their business impact.

Key Problems to Address:

- Inventory inaccuracies leading to picking errors and delays.
- Inefficient picking routes increasing time and labor costs.
- Lack of real-time visibility and updates between warehouse floor and WMS.
- Manual processes causing human errors and poor traceability.

Importance:

• Explain why solving these issues is critical for operational efficiency, cost reduction, and customer satisfaction.

2. Target Users

Primary Users:

- **Pickers:**
  - Responsible for physically retrieving items from warehouse locations.
  - Needs: speed, accuracy, easy-to-use interface, minimal manual entry.
- **Warehouse Supervisors:**
  - Oversee picking operations, monitor performance, handle exceptions (e.g., substitutions, cycle counts).

    - Needs: visibility into operations, ability to resolve issues, access to real-time data.

Secondary Users:

- IT/Admin: Manage user access, troubleshoot integration issues.

3. User Flow / App Flow

Step-by-Step Flow:

1. **Login & Authentication:**
    - Secure sign-in using employee credentials or badge scan.
2. **Picklist Selection:**
    - View list of assigned picklists/orders with priority indicators.
3. **Optimized Navigation:**
    - App displays the most efficient route through the warehouse (map or step-by-step directions).
4. **Item Picking:**
    - For each item:
        - View item details (image, SKU, quantity, bin location).
        - Scan barcode to confirm pick.
        - Mark as “Picked” or “Not Available.”
5. **Exception Handling:**
    - If item not found:
        - App suggests alternative location/SKU.
        - Option to trigger a “cycle count” for inventory verification.
6. **Completion & Sync:**
    - After all items are picked, user reviews and confirms completion.
    - Data is synced with WMS, updating order and inventory status.

4. Key Features

- **Barcode Scanning:** Fast, accurate item validation.
- **Route Optimization:** Shortest picking path based on bin locations.
- **Offline Mode:** Continue work without connectivity; auto-sync when online.
- **Item Substitution:** Suggests alternatives when items are unavailable.
- **Real-Time WMS Sync:** Immediate updates to inventory and order status.
- **Exception Reporting:** Log issues (e.g., missing items, substitutions, cycle counts).
- **User-Friendly Interface:** Large buttons, clear visuals, glove-friendly design.
- **Audit Trail:** Tracks all actions for accountability.

5. Integration Points

WMS Communication:

- **APIs:**
    - Fetch picklists, inventory levels, bin locations.
    - Push pick confirmations, substitutions, cycle count triggers.
- **Webhooks/Real-Time Events:**
    - Notify WMS of pick completion or exceptions.
- **Data Sync:**
    - Bi-directional, scheduled or real-time sync to ensure data consistency.

Security & Access:

- Role-based access control, secure data transmission.

6. Sample UI

https://www.figma.com/proto/rUvvFzIm4sJG4i4sdRAQV6/Untitled?node-id=0-1&t=9GNV8osji8mPjoPR-1

7. Success Metrics (KPIs)

- **Pick Accuracy Rate:** % of items picked correctly on first attempt.
- **Average Picklist Completion Time:** Time taken from start to finish per picklist.
- **Cycle Count Trigger Rate:** % of picks requiring inventory verification.
- **Order Fulfillment Time:** Time from order assignment to ready-for-shipment.
- **Substitution Rate:** % of items substituted due to stockouts.